MIDTERM EXAMS SEPTEMBER 2025

III SEM Aeronutical Engineering

MATERIALS AND PROCESSING TECHNIQUES [AAE 2122]

ANSWER KEY

Type: MCQ

Q1. The degree of freedom at triple point in unary diagram for water _____ (0.5)

1. **** 0**
2. 1
3. 2
4. 3

Q2. Following is wrong about a phase diagram (0.5)

1. ****It gives information on transformation rates**
2. Relative amount of different phases can be found under given equilibrium conditions.
3. It indicates the temperature at which different phases start to melt.
4. Solid solubility limits are depicted by it.

Q3. Not a Hume-Ruthery condition (0.5)

1. Crystal structure of each element of solid solution must be the same
2. ****Elements should form compounds with each other**
3. Elements should have the same valence
4. Size of atoms of each two elements must not differ by more than 15%

Q4. The boundary line between (alpha) and (alpha+beta) regions must be part of _____ (0.5)

1. Solidus
2. Liquidus
3. ****Solvus**
4. Tie-line

Q5. In a binary eutectic system, just above eutectic temperature, the phases present are (0.5)

1. α + β
2. L + α

3. **Only Liquid
4. L + β

Q6. Which one represents a family of equivalent planes? (0.5)

1. **{hkl}
2. (hkl)
3. [uvw]
4. ⟨uvw⟩

Q7. In cubic crystals, direction [200] is equivalent to (0.5)

1. [110]
2. **[100]
3. [111]
4. [120]

Q8. Edge dislocation is characterized by (0.5)

1. Screw deformation.
2. **Extra half-plane of atoms.
3. Vacancy cluster.
4. Interstitial impurity.

Q9. Burgers vector is perpendicular to dislocation line in (0.5)

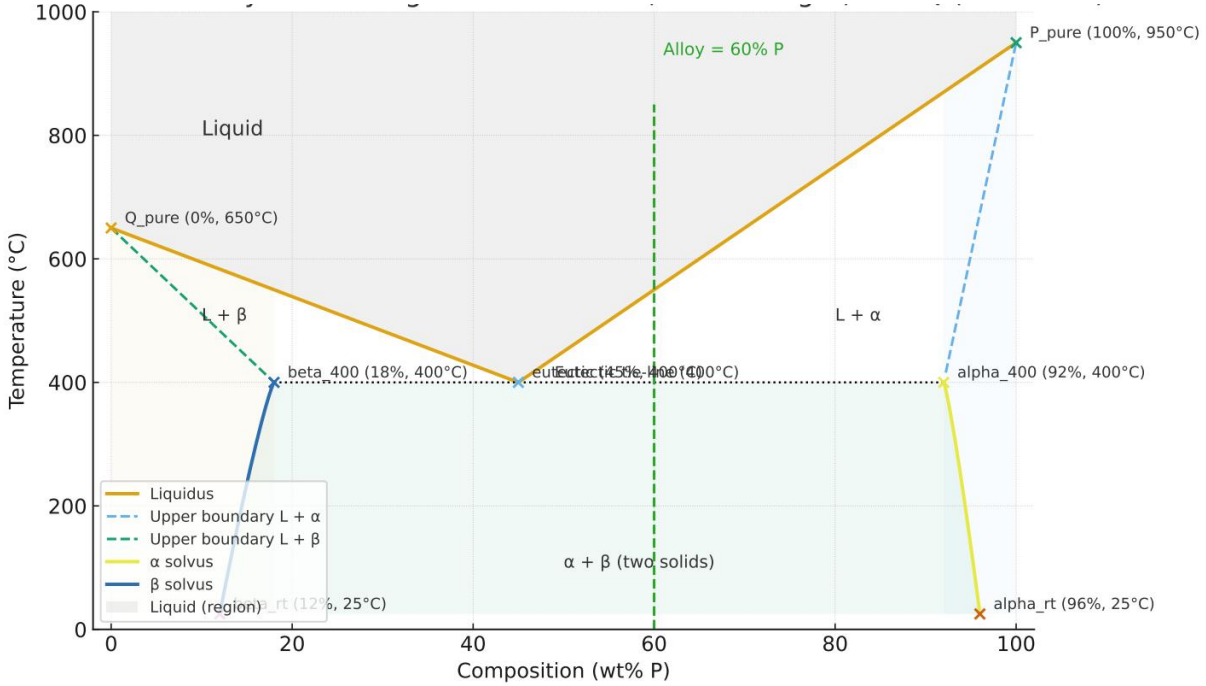
1. Screw dislocation
2. **Edge dislocation
3. Mixed dislocation
4. Both screw and edge

Q10. Ductility of material is measured by (0.5)

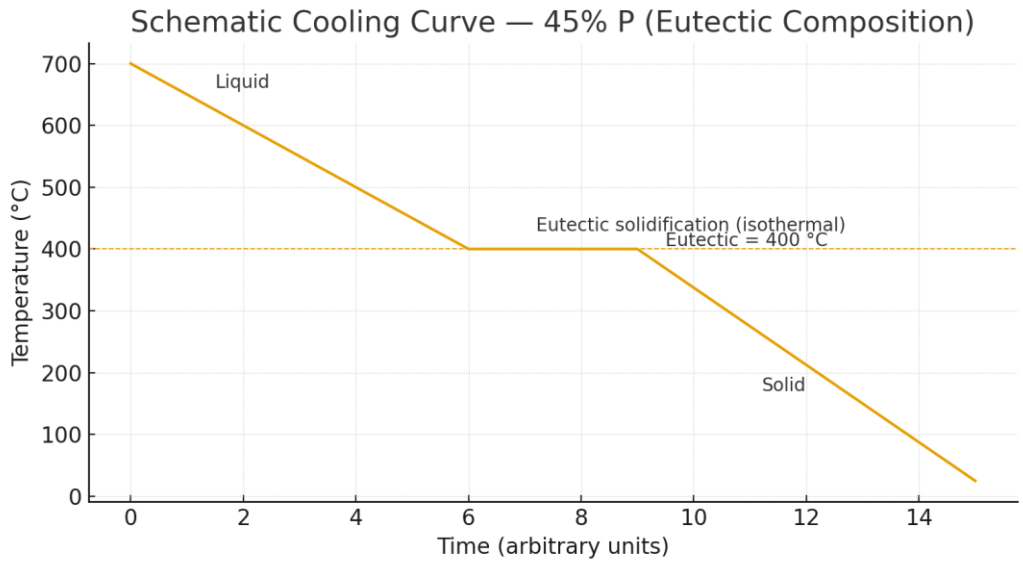
1. **Elongation %
2. Brinell hardness test
3. Impact test
4. Modulus of elasticity

Type: DES

Q11. Consider an alloy system of Metals P and Q with melting points of 950 C and 650 C. A eutectic reaction occurs at 400 C with a composition of 55% Q. The maximum solid solubility of P in Q is 18% at 400 C, and P in Q is 12% at room temperature. The maximum solid solubility of Q in P is 8% at 400 C, and 4% at room temperature. For a 60% P alloy, determine: a) The mass fraction of the primary solid phase just above the eutectic temperature. b) The mass fraction of the eutectic structure just below the eutectic temperature. c) The temperature at which solidification is 50% complete. d) Sketch the cooling curve for a 45% P alloy. (4)



Sketch the cooling curve for a 45% P alloy



For the two-phase region (L + α) at 400 °C the lever rule gives:

$$f_\alpha = \frac{C_0 - C_L}{C_\alpha - C_L}$$

where C_L is the liquid composition on the tie-line at the eutectic (here $C_L = C_E = 45\%$).

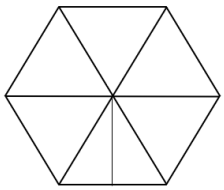
Substitute numbers:

$$f_\alpha = \frac{60 - 45}{92 - 45} = \frac{15}{47} \approx 0.31915$$

Answer (a): $f_\alpha \approx 0.319$ (31.9%).

$f_{eutectic} \approx 0.681$ (68.1%).

Q12. Derive the APF expression for HCP structure. (3)



Area of HCP basal plane

The volume of the hexagonal unit cell,

V= Area of the hexagonal basal plane × height

The hexagonal basal plane can be divided into six equilateral triangles as shown in fig.

Then, area of the hexagonal basal plane = 6 × area of triangle PQR

Altitude of the triangle PQR, h = tan 60° × a/2 = $\frac{a\sqrt{3}}{2}$

Area of triangle PQR = ½ × a × h = ½ × a × $\frac{a\sqrt{3}}{2}$ = $\frac{a^2\sqrt{3}}{4}$

Area of the hexagonal basal plane = **6** × $\frac{a^2\sqrt{3}}{4}$ = $\frac{3\sqrt{3}a^2}{2}$

Volume of the hexagonal unit cell, V = Area of the hexagonal basal plane × height

Height of HCP unit cell, $c = a\sqrt{\frac{8}{3}} = 1.633a$

$\therefore$ V = $\frac{3\sqrt{3}a^2}{2} \times a\sqrt{\frac{8}{3}} = 3\sqrt{2}a^3$

It may be observed that, a = 2r

$$\eta_{HCP} = \frac{N_{HCP} \times \left(\frac{4\pi r^3}{3}\right)}{V} \times 100$$

$$= \frac{N_{HCP} \times \left(\frac{4\pi r^3}{3}\right)}{3\sqrt{2}a^3} \times 100$$

$$= \frac{6 \times \left(\frac{4\pi r^3}{3}\right)}{3\sqrt{2} \times (2r)^3} \times 100 = \frac{24\pi r^3}{3} \times \frac{1}{24\sqrt{2}r^3} \times 100$$

$$= 74.05\%$$

Q13. Define twinning. Differentiate between slip and twinning. (3)

Twinning is a mode of plastic deformation in crystals in which a portion of the crystal lattice undergoes a homogeneous shear such that it forms a mirror image of the original lattice across a specific crystallographic plane, called the twin plane.

- In twinning, the orientation of atoms changes in a well-defined, coordinated manner, unlike random displacements.
- It is commonly observed in HCP and BCC metals where slip systems are limited.

Aspect	Slip	Twinning
Definition	Plastic deformation by relative motion of one part of the crystal over another along a specific crystallographic plane (slip plane).	Plastic deformation where a portion of the crystal becomes a mirror image of the other part across a twin plane.
Mechanism	Involves **dislocation motion** → atoms move by one interatomic spacing at a time.	Involves **homogeneous shear** → atoms shift cooperatively to a new orientation.
Plane involved	Occurs on the most densely packed plane (slip plane).	Occurs on a specific crystallographic twin plane.
Extent of deformation	Produces **large strains** since dislocations can move repeatedly over long distances.	Produces only **small strains** (limited reorientation of lattice).
Atomic movement	Atoms move over many interatomic distances.	Atoms move a fraction of interatomic distance in a coordinated manner.
Result	No change in lattice orientation, just relative displacement.	Change in crystal orientation (new orientation is a mirror image across twin plane).
Occurrence	Dominant in FCC metals (where many slip systems are available).	More common in HCP and BCC metals, especially at low temperatures and high strain rates.

Q14. Explain vacancy diffusion and interstitial diffusion mechanisms. (3)

Vacancy diffusion occurs when an atom moves from its lattice site to a neighboring vacant site (vacancy).

Mechanism:

1. A lattice site becomes vacant (due to thermal vibrations or imperfections).
2. A neighboring atom jumps into this vacancy.
3. The process repeats, effectively transporting atoms through the lattice.

Interstitial Diffusion

- Interstitial diffusion occurs when small atoms (like H, C, N, O) move through the interstitial sites (spaces between larger host atoms) in the crystal lattice.
- Mechanism:
    1. Interstitial atoms occupy small voids (tetrahedral or octahedral sites).
    2. These atoms move rapidly from one interstitial site to another.

Q15. Explain recovery, recrystallization, and grain growth processes. (3)

Recovery

- Recovery is the process that occurs at relatively low temperatures (below recrystallization temperature) where the stored internal strain energy in the cold-worked metal is partially relieved without major changes in microstructure

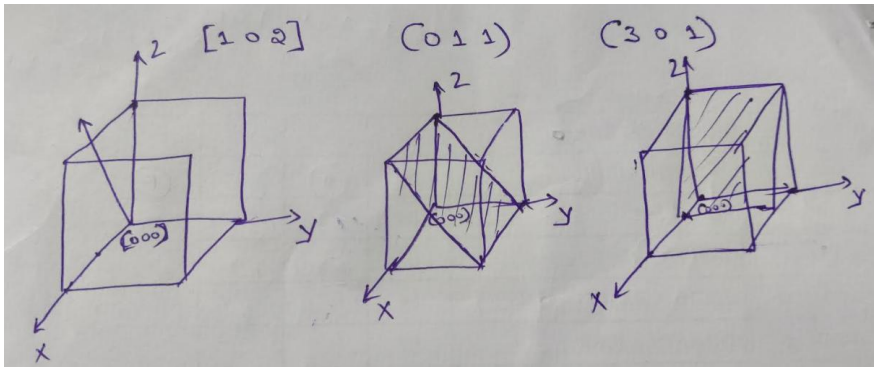
Recrystallization

- Recrystallization is the process where new strain-free grains nucleate and grow within the deformed microstructure, replacing the old work-hardened grains.

Grain Growth

- Grain growth occurs if recrystallized metal is kept at elevated temperature for longer time, where larger grains grow at the expense of smaller ones.

Q16. Show the miller indices for the following [1 0 2], (0 1 1), (3 0 1) . (3)



Q17. What is the significance of tie-lines in a eutectic phase diagram?. (2)

Significance:

- A tie-line connects the boundaries of the two phases in equilibrium.
- The intersection of the tie-line with the phase boundaries gives the composition of each phase present at that temperature.
- Example: In the (α + β) region, the left end of the tie-line gives the composition of α-phase**,** and the right end gives that of β-phase**.**

Q18. Write any four criteria for selection of materials in engineering design. (2)

1. Strength, stiffness, hardness, toughness, fatigue resistance, creep resistance.
2. Ensures the material can withstand applied loads and stresses without failure.
3. Density, thermal conductivity, electrical conductivity, corrosion resistance, oxidation resistance.
4. Important for performance in different service environments.
5. Machinability, castability, weldability, formability, heat treatability.
6. Material should be compatible with the chosen manufacturing processes.
7. Cost of raw material, processing cost, ease of sourcing, and availability.
8. Critical for ensuring overall project feasibility and sustainability.

Q19. List the Hume-Rothery rule. (2)

Atomic Size Factor (Size Difference Rule)

- The difference in atomic radii of solute and solvent should be less than 15%.
- If the difference is larger, lattice distortion becomes too high, and solubility is limited.

Crystal Structure Rule

- The two metals should have the same crystal structure (FCC, BCC, HCP, etc.) to have high solubility.
- If crystal structures differ, solubility is very limited.

Electronegativity Rule

- The electronegativity difference between the two metals should be small.
- If it is large, the metals tend to form intermetallic compounds rather than a solid solution.

Valency Rule

- A metal tends to dissolve another metal of higher valency more readily than one of lower valency.
- Solvent with higher valency dissolves more solute atoms.